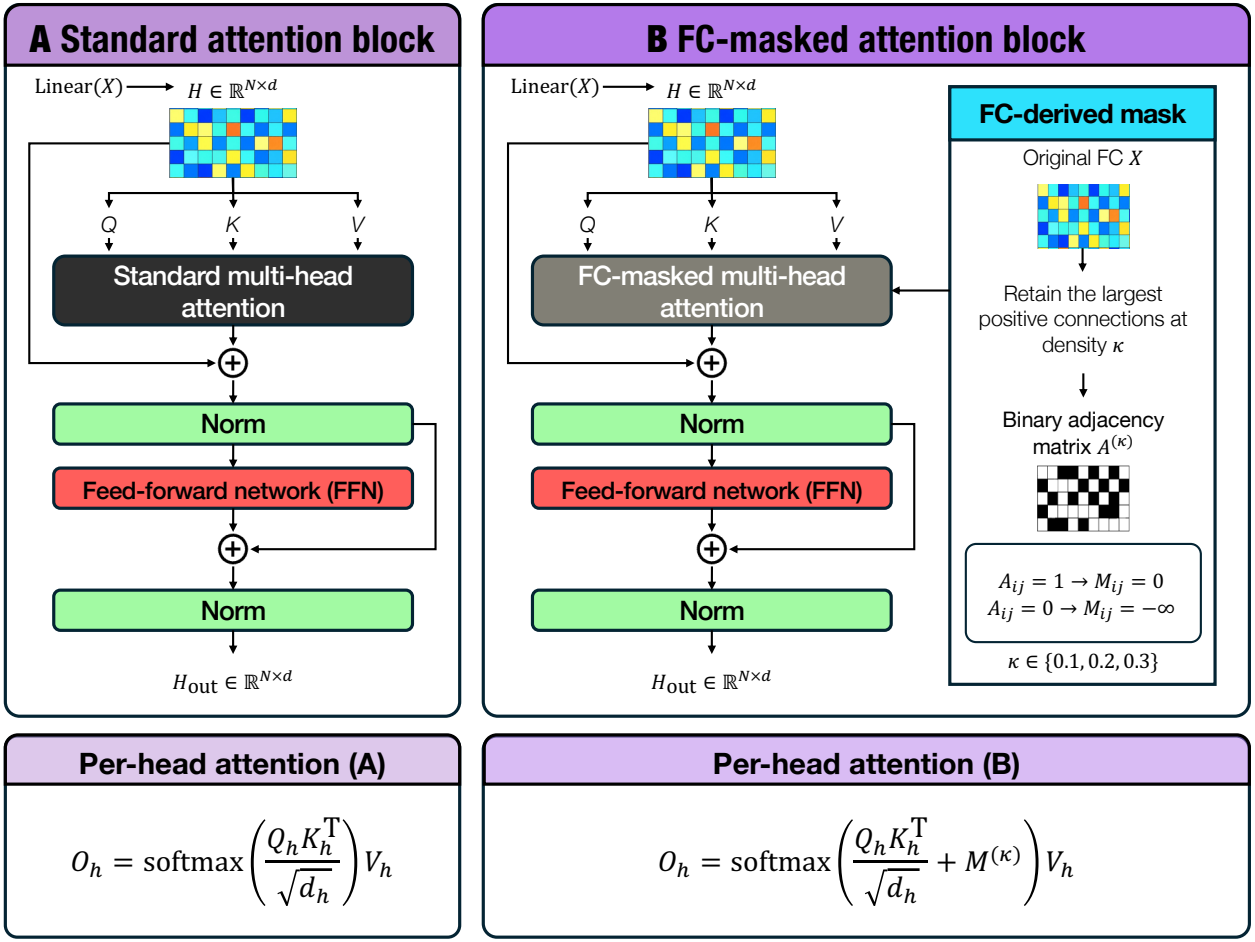

A Standard attention block
Linear(X)
H ∈ ℝ^{N×d}
Q
K
V
Standard multi-head attention
Norm
Feed-forward network (FFN)
Norm
H_out ∈ ℝ^{N×d}
B FC-masked attention block
Linear(X)
H ∈ ℝ^{N×d}
Q
K
V
FC-masked multi-head attention
Norm
Feed-forward network (FFN)
Norm
H_out ∈ ℝ^{N×d}
FC-derived mask
Original FC X
Retain the largest positive connections at density κ
Binary adjacency matrix A^(κ)
A_ij = 1 → M_ij = 0
A_ij = 0 → M_ij = −∞
κ ∈ {0.1, 0.2, 0.3}
Per-head attention (A)
O_h = softmax(Q_h K_h^T / √d_h) V_h
Per-head attention (B)
O_h = softmax(Q_h K_h^T / √d_h + M^(κ)) V_h